A HYBRID APPROACH FOR IMAGE ENHANCEMENT AND VISIBILITY RESTORATION UNDER ATMOSPHERIC DEGRADATION

A Project report submitted in partial fulfillment of the requirements for the

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ELECTRONICS & COMMUNICATION ENGINEERING

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We certify that the work contained in this report is original and has been done by us under the guidance of my supervisor. The work has not been submitted to any other Institute for any degree or diploma. We have followed the guidelines provided by the Institute in preparing the report. we have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

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CO4	Apply Communication & Presentation skills	Apply
CO5	Develop the team work, leadership skills with professional and ethical values.	Understand
CO6	Make the use of report writing skills	Understand

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ABSTRACT

Image visibility degradation caused by atmospheric conditions such as fog, smoke, and sand Dust significantly affects the quality of captured images by reducing contrast, distorting colors, and hiding important scene details. These degradations pose challenges for various computer vision applications including surveillance systems, intelligent transportation, remote sensing, and outdoor monitoring. The clear image visibility is crucial for accurate analysis and decision-making. To address this issue, our proposed work presents a visibility restoration and image enhancement method for images degraded by atmospheric scattering conditions. Our novel approach is based on the combination of Dark Channel Prior(DCP) technique, with contrast limited adaptive histogram equalization(CLAHE) widely helpful for single-image de-hazing. In this method, the degraded input image is first processed to compute the dark channel, which represents the minimum intensity value within a local image patch and helps estimate the density of haze in the scene. Using the dark channel information, atmospheric light is estimated from the brightest pixels of the dark channel map. Then transmission map is calculated to determine the amount of scene radiance that reaches the camera through the degraded medium. This transmission map is further refined using filtering techniques to obtain smoother and more accurate results. Based on the refined transmission map and estimated atmospheric light, the scene radiance is recovered using the atmospheric scattering model to remove haze effects. Finally, CLAHE is applied to improve image clarity and color distribution. Our proposed results demonstrate effectively to enhance the image visibility which are evaluated by peak signal to noise ratio(PSNR) and structural similarity index measure(SSIM). Among these results, our proposed method shows better clarity as well as in metrics and improved scene details in images affected by fog, smoke, and sand dust conditions.

Keywords: Image Enhancement, Visibility Restoration, Haze removal, Dark Channel Prior, Fog, Smoke, Sand Dust, Transmission Map, Atmospheric Scattering Model.

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LIST OF ABBREVIATIONS

Abbreviations	Descriptions
IOT	: Internet of Things
MPA	: Microstrip Patch Antenna
BW	: Bandwidth
WBAN	: Wireless Body Area Networks
HFSS	: High-Frequency Structure Simulator
WPT	: Wireless Power Transfer
WSN	: Wireless Sensor Networks
RF	: Radio Frequency
GSM	: Global System for Mobile Communications
LTE	: Long-Term Evolution
2G	: Second Generation
3G	: Third Generation
4G	: Fourth Generation
5G	: Fifth Generation
VSWR	: Voltage Standing Wave Ratio
RL	: Return Loss
EH	: Energy Harvesting
GP	: Ground Plane
EM	: Electromagnetic
PD	: Power Density
PTE	: Power Transmission Efficiency

CHAPTER

INTRODUCTION

1.1 Introduction

Image processing plays a vital role in modern technology by enabling the analysis, enhancement, and interpretation of visual data. With the rapid growth of computer vision systems, the quality of input images has become increasingly important for accurate decision-making in various applications. However, in real-world scenarios, images are often captured under unfavorable environmental conditions that significantly degrade their quality. Among these conditions, atmospheric phenomena such as fog, smoke, and sand dust are major factors that reduce image clarity and visibility.

These atmospheric disturbances affect image formation due to the interaction between light and particles present in the air. When light travels from an object to a camera, it gets scattered and partially absorbed by these particles. This scattering causes deviation of light rays, while absorption reduces the intensity of light reaching the camera. As a result, the captured image suffers from reduced contrast, blurred details, color distortion, and overall poor visibility. Such degradation not only affects human perception but also reduces the performance of automated systems that rely on visual input.

Fog is one of the major causes of image degradation. It is formed by tiny water droplets suspended in the air, which scatter light in different directions. Due to this scattering effect, images captured in foggy conditions appear faded and unclear, as shown in Figure 1.1. The contrast of the image decreases significantly, and distant objects become less visible compared to nearby ones. In addition, colors lose their brightness and shift toward gray or white tones. This makes object detection and recognition difficult, especially in applications such as traffic monitoring, surveillance systems, and autonomous vehicles.

Figure 1 . 3 : Fog Image

Smoke is another atmospheric condition that affects image quality, especially in areas with fire, industrial pollution, or vehicle emissions. Smoke consists of gases and fine particles produced during combustion. Unlike fog, smoke is irregular and non-uniform in nature, which makes image degradation more complex, as illustrated in Figure 1.2. It causes uneven visibility across the image, where some areas may appear clear while others are heavily obscured. Smoke also introduces blurring, noise, and color distortion, often making the image appear darker or grayish. Moreover, smoke is dynamic and changes over time, which makes enhancement more difficult, particularly in video processing. Removing smoke effects while preserving image details is a challenging task, but it is essential in applications such as fire detection, rescue operations, and industrial safety monitoring.

Figure 1.2: Smoke Image

Sand dust is another significant factor that causes severe image degradation, especially in desert and dry regions. It occurs when strong winds carry sand particles into the air, forming dust storms. These particles are larger than those in fog and smoke, leading to stronger scattering and absorption of light. As shown in Figure 1.3, images captured in such conditions have very low visibility and suffer from heavy color distortion. The images often appear yellowish or reddish due to the natural color of sand particles.

Figure 1 . 3 : Sand Dust Image

Important details and textures are lost, and in extreme cases, objects may become almost invisible. Enhancing sand dust images is more difficult compared to fog and smoke because of the severe loss of information and strong color bias. It requires advanced techniques to restore both visibility and natural color balance.

To overcome these problems, various image enhancement techniques are used. One of the most effective methods for removing haze from images is the Dark Channel Prior (DCP). This method is based on the observation that in most natural images, at least one color channel contains very low intensity values in non-sky regions. Using this property, the DCP algorithm estimates the thickness of haze present in the image and recovers the original scene by reducing the effect of atmospheric scattering. It is widely used for dehazing foggy and smoky images due to its effectiveness in restoring visibility.

In addition to haze removal, improving contrast is also necessary to enhance image quality. For this purpose, our project showcase some enhancement method such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and Dark Channel Prior (DCP). CLAHE enhances the local contrast of the image and brings out hidden details without amplifying noise excessively. It works by dividing the image into small regions and applying histogram equalization to each region, ensuring better enhancement results. We are using a combination of DCP and CLAHE provides an effective solution for enhancing images affected by fog, smoke, and sand dust. While DCP focuses on removing haze and restoring visibility, CLAHE improves contrast and detail representation. Together, they produce clearer and more natural-looking images.

1.2 Motivation of Thesis

The main motive of this project is to address the problem of poor image quality caused by atmospheric conditions such as fog, smoke, and sand dust. In many real-world situations, images captured in outdoor environments suffer from reduced visibility, low contrast, and loss of important details due to the scattering and absorption of light by particles present in the air. These degraded images make it difficult for both humans and computer-based systems to analyze and interpret visual information accurately. With the increasing use of image-based technologies in fields such as transportation, surveillance, and environmental monitoring, there is a growing need for clear and high-quality images. For example, in traffic systems and autonomous vehicles, poor visibility due to fog or dust can lead to serious accidents. Similarly, in fire incidents, smoke can obstruct vision and make rescue operations more difficult. These

real-world challenges motivate the need for effective image enhancement techniques that can improve visibility and restore useful information from degraded images.

Another important motive of this project is to improve the performance of computer vision systems. Many modern applications depend on accurate image analysis for tasks such as object detection, recognition, and tracking. When images are affected by atmospheric conditions, the performance of these systems decreases significantly. By enhancing image quality, the reliability and efficiency of such systems can be greatly improved. This project is also motivated by the effectiveness of advanced image processing techniques such as Dark Channel Prior (DCP) and Contrast Limited Adaptive Histogram Equalization (CLAHE). These methods provide efficient ways to remove haze and improve contrast in degraded images. By combining these techniques, the project aims to develop a practical solution for enhancing images affected by fog, smoke, and sand dust. Thus, the overall motive of this project is to develop an efficient image enhancement approach that improves visibility, restores important details, and produces clear images that can be effectively used in real- world applications.

1.3 Objective of Thesis

The primary objective of this thesis is to develop an effective image enhancement system for improving the quality of images degraded by atmospheric conditions such as fog, smoke, and sand dust. The proposed work focuses on enhancing visibility, contrast, and detail preservation using advanced image processing techniques.

The specific objectives of this research are as follows:

- To analyze the effects of atmospheric conditions such as fog, smoke, and sand dust on image quality, including contrast reduction, color distortion, and loss of fine details.
- To implement the Dark Channel Prior (DCP) method for haze removal and atmospheric scattering estimation in order to recover clear and visually improved images.
- To apply Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance local contrast and reveal hidden details in degraded images.

- To integrate DCP and CLAHE techniques to achieve superior enhancement results compared to the application of individual methods.
- To evaluate the performance of the proposed enhancement system based on qualitative parameters such as PSNR and SSIM for visibility improvement and visual quality.
- To validate the effectiveness of the proposed approach using various degraded images affected by fog, smoke, and sand dust conditions.
- To develop a simple, efficient, and practical system that can be applied in real-world applications such as surveillance systems, autonomous driving, and environmental monitoring.

This thesis aims to develop an efficient and reliable image enhancement approach for improving the quality of images affected by fog, smoke, and sand dust. By integrating Dark Channel Prior (DCP) and Contrast Limited Adaptive Histogram Equalization (CLAHE), the proposed method focuses on achieving better visibility, enhanced contrast, and restoration of important details, making the images more suitable for practical applications such as different climate conditions, medical imaging, surveillance, remote sensing, and computer vision.

1.4 Organization of Thesis

This thesis is organized into seven chapters to systematically present the work carried out on image enhancement of fog, smoke, and sand dust affected images using the proposed method.

Chapter 1: Introduction

This chapter provides an overview of the project, including background information on image degradation due to fog, smoke, and sand dust. It explains the need for enhancement techniques and presents the motivation, objectives, and scope of the thesis clearly.

Chapter 2: Literature Review

This chapter discusses various existing image enhancement and dehazing techniques proposed by different researchers. It highlights their previous working principles, advantages, limitations, and the need for improved methods to handle complex atmospheric conditions

effectively.

Chapter 3: Introduction to Image Processing

This chapter introduces the basic concepts of image processing, including image representation, types of images, and fundamental operations. It also explains common enhancement techniques and their importance in improving image quality for analysis.

Chapter 4: MATLAB Software

This chapter provides an overview of MATLAB, including its features, tools, and image processing capabilities. It explains how MATLAB is used for implementing algorithms and why it is suitable for this project.

Chapter 5: Proposed Methodology

This chapter explains the proposed approach for image enhancement using the Dark Channel Prior (DCP) method. It describes each step of the algorithm, including haze estimation, transmission calculation, and image restoration process.

Chapter 6: Results and Discussion

This chapter presents the results obtained after applying the proposed method on various degraded images. It includes analysis of visibility improvement, contrast enhancement, and comparison with original images to evaluate overall performance.

Chapter 7: Conclusion and Future Scope

This chapter summarizes the entire work carried out in the thesis and highlights the key outcomes. It also discusses possible future improvements, such as combining additional techniques to further enhance image quality and performance.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Survey

Previous studied research papers which related to image dehazing techniques, focusing on methods such as Dark Channel Prior (DCP), Non-Local Filtering, and K-SVD based approaches. The need for effective dehazing arises due to atmospheric conditions like fog, smoke, and sand dust, which degrade image quality by reducing visibility, contrast, and color accuracy. Among the existing techniques, the DCP method is widely used due to its ability to estimate transmission maps and atmospheric light for restoring clear images. Non-local filtering methods improve image quality by utilizing redundancy in similar patches, while K-SVD based methods focus on sparse representation for effective noise removal and detail preservation. This study provides a comparative analysis of these methods to identify their strengths and limitations, forming the basis for the proposed dehazing approach.

Wang and Durmus (2025) studied the effect of color contrast on the visual clarity of images in complex indoor environments. Their work emphasizes that visual perception plays a crucial role in image quality assessment, especially in scenes with multiple objects and varying lighting conditions. The authors analyzed how adjusting color contrast can significantly improve the visibility and distinguishability of objects without altering the structural content of the image. They conducted experiments on indoor images and observed that proper contrast enhancement leads to better human perception and scene understanding. However, the study mainly focuses on perceptual improvement rather than physical restoration of degraded images. It does not directly address atmospheric degradation such as fog, smoke, or sand dust. Despite this limitation, the findings are useful in post-processing stages of image dehazing, where contrast enhancement is required to refine the final output. Thus, this work provides a foundation for improving visual quality after applying dehazing algorithms [1].

Cheng *et al*. (2020) proposed a fast and efficient sand-dust image enhancement algorithm based on blue channel compensation and guided image filtering. The study identifies that sand-dust degraded images suffer from severe color distortion, particularly in the blue channel, due to atmospheric scattering effects. To address this issue, the authors enhance the blue channel intensity

and apply guided filtering to refine the image while preserving edges and important structural details. The proposed method improves both color balance and visibility, producing more natural-looking images. One of the key advantages of this approach is its low computational complexity, making it suitable for real-time applications. Experimental results demonstrate that the method outperforms traditional enhancement techniques in terms of clarity and color restoration. However, the method may not perform well in dense haze conditions where depth estimation is required. This approach can be effectively combined with dehazing techniques like Dark Channel Prior for improved performance [2].

Masood *et al*. (2025) introduced an advanced sand-dust image enhancement technique using intensity-corrected blue channel compensation combined with edge-preserving contrast enhancement. The authors improved upon existing blue channel methods by incorporating Laplace filtering and a sigmoid function to enhance edges and control contrast effectively. The proposed method aims to restore image details while maintaining natural color balance and avoiding over-enhancement. Experimental results show that the technique significantly improves visibility and preserves fine details compared to traditional methods. Additionally, the approach reduces noise and enhances structural information in degraded images. However, the inclusion of multiple processing steps increases computational complexity, making it less suitable for real-time applications. The method is particularly effective for heavily degraded sand-dust images and can complement dehazing algorithms such as Dark Channel Prior. Overall, this work contributes to improving both color correction and edge preservation in challenging for the environmental conditions [3].

Santos *et al*. (2025) proposed a discrete spatial diffusion model for image restoration, focusing on preserving image intensity while enhancing quality. The method is based on diffusion principles, where pixel values are adjusted iteratively to reduce noise and improve smoothness without affecting important features. The authors emphasized maintaining the original intensity distribution, which is often distorted in traditional enhancement methods. Experimental results demonstrate that the model effectively improves image quality while preserving structural details. Although the method is not specifically designed for atmospheric degradation, it can handle various types of image distortions. The approach is particularly useful for refining images after initial restoration, such as dehazing. However, it lacks specific mechanisms to estimate atmospheric parameters like transmission maps. Despite this limitation, the diffusion-based approach provides a valuable contribution to image enhancement and can be integrated with dehazing methods to achieve better results [4].

Singh *et al.* (2022) proposed a fusion-based visibility restoration model for hazy remote sensing images using dynamic differential evolution. The main objective of this work is to improve image clarity by combining multiple restoration techniques through an optimization framework. The authors utilized differential evolution to dynamically select optimal parameters, enhancing the effectiveness of the fusion process. This approach enables better handling of varying haze densities and improves both contrast and detail preservation. Experimental results demonstrate that the fusion-based model performs better than single-method approaches, especially in complex environmental conditions. However, the method involves high computational complexity due to the optimization process and multiple stages of processing. It may not be suitable for real-time applications. Despite this limitation, the study highlights the importance of combining multiple techniques for effective image restoration. This approach provides useful insights for integrating optimization strategies with dehazing methods such as Dark Channel Prior [5].

Liang *et al*. (2022) introduced a generalized image formation model (GIFM) for restoring images captured under poor visibility conditions. The proposed method extends traditional atmospheric scattering models by incorporating additional parameters to handle various types of degradation, including haze, fog, and dust. The model estimates degradation factors more accurately, resulting in improved restoration performance. Experimental results show that the method enhances image clarity, contrast, and color fidelity compared to conventional approaches. One of the key strengths of this work is its adaptability to different environmental conditions. However, accurate estimation of model parameters remains a challenge, especially in real-world scenarios where conditions vary significantly. The method also involves complex computations, which may limit its practical implementation. Nevertheless, this study provides a strong theoretical foundation for image dehazing and can be compared with Dark Channel Prior to evaluate performance under different degradation conditions [6].

Muthuraman and Santhanam (2022) focused on improving visibility in underwater turbid images using a hybrid restoration network combined with weighted filtering. The proposed method integrates deep learning techniques with traditional filtering approaches to enhance image quality. The weighted filter helps in reducing noise while preserving edges and important features. Experimental results indicate significant improvements in clarity, contrast, and color correction. Although the method is specifically designed for underwater images, the degradation characteristics are similar to atmospheric conditions such as fog and

smoke, where scattering and absorption affect image quality. Therefore, the approach can provide valuable insights for atmospheric image dehazing. However, the model requires a large amount of training data and computational resources, making it less efficient for real-time applications. This work demonstrates the effectiveness of hybrid models in handling complex degradation scenarios and can be compared with traditional methods like DCP [7].

Li *et al*. (2020) proposed a hybrid framework for underwater image enhancement by combining multiple image processing techniques. The method focuses on improving visibility, contrast, and color balance in degraded underwater images. The authors utilized a combination of enhancement algorithms to address different aspects of image degradation, resulting in improved overall performance. Experimental results show that the proposed framework produces better visual quality compared to single-method approaches. Although the method is designed for underwater conditions, it shares similarities with atmospheric degradation such as fog and sand dust. The study highlights the importance of combining multiple techniques for effective image enhancement. However, the approach does not explicitly estimate atmospheric parameters like transmission maps, which are important in dehazing. Despite this limitation, the hybrid framework provides useful insights for developing advanced image restoration techniques and can complement methods like Dark Channel Prior [8].

Yasarla and Patel (2022) presented a convolutional neural network (CNN)-based approach for restoring images degraded by atmospheric turbulence. The proposed method learns complex degradation patterns directly from data and reconstructs high-quality images from distorted inputs. The model effectively reduces distortions and improves image clarity and sharpness. Experimental results demonstrate that the CNN-based approach outperforms traditional image restoration methods. However, the method requires large training datasets and high computational power, which can be a limitation in practical applications. Although the study focuses on atmospheric turbulence, the approach can be extended to other types of degradation such as fog and haze. This work highlights the growing importance of deep learning in image restoration. Compared to traditional methods like Dark Channel Prior, CNN-based methods offer higher accuracy but at the cost of increased complexity and resource requirements [9].

Cheng *et al*. (2023) proposed a CNN-based method for restoring short-exposure images affected by atmospheric turbulence. The model is designed to reconstruct clear images by learning the mapping between degraded and high-quality images. The approach improves

image sharpness, reduces noise, and enhances structural details. Experimental results show that the method achieves better performance compared to conventional restoration techniques. However, the model requires extensive training and may not generalize well to all types of degradation conditions. The study emphasizes the effectiveness of deep learning in handling complex image distortions. Although the focus is on turbulence, the approach can be adapted for dehazing applications. Compared to traditional methods like Dark Channel Prior, this method provides improved accuracy but requires more computational resources. This work contributes to the advancement of learning-based different image restoration techniques [10].

Zhu *et al*. (2024) introduced MWFormer, a transformer-based model for multi-weather image restoration. The method utilizes attention mechanisms to handle different types of degradation, including haze, rain, and dust. The model adapts to varying environmental conditions and provides high-quality restoration results. Experimental studies show that the transformer-based approach outperforms CNN-based models in terms of accuracy and detail preservation. However, the model is computationally expensive and requires large datasets for training. This limits its practical applicability in resource- constrained environments. The study represents a significant advancement in deep learning- based image restoration. Compared to traditional methods like Dark Channel Prior, MWFormer offers improved performance but at the cost of higher complexity. This work highlights the trend toward using advanced neural network architectures for image enhancement [11].

Siddiqua *et al*. (2023) proposed MACGAN, a GAN-based model for image restoration under multiple adverse conditions. The method uses a multidomain attention mechanism to enhance image quality and restore fine details. The GAN framework generates high-quality images by learning from degraded inputs. Experimental results show that the method effectively improves contrast, sharpness, and overall visual quality. However, GAN-based models are complex and require careful training to avoid issues such as instability and artifacts. The approach demonstrates the effectiveness of generative models in image restoration. Compared to traditional methods like Dark Channel Prior, GAN-based approaches provide better visual quality but involve higher computational complexity. This work contributes to the development of advanced learning-based restoration techniques [12].

Dong *et al*. (2026) proposed a physics-guided degradation-adaptive model for remote sensing image restoration. The method combines physical modeling with deep learning to improve restoration accuracy. The model adapts to different degradation conditions by learning from both physical principles and data. Experimental results show improved clarity,

contrast, and detail preservation. However, the method involves complex computations and requires parameter tuning. The integration of physics-based and learning-based approaches makes this method robust and effective. Compared to traditional methods like Dark Channel Prior, this approach provides better adaptability but at the cost of increased complexity. This work highlights the importance of hybrid models in image restoration [13].

Gao *et al.* (2026) introduced a lightweight dual-view network for sand-dust image enhancement. The method uses dual-view inputs to improve restoration performance while maintaining computational efficiency. The approach is designed for real-time applications and produces improved image clarity and reduced distortion. Experimental results demonstrate that the method achieves good performance with lower computational cost compared to deep learning models. However, it may not perform well under extreme degradation conditions. This work focuses on balancing performance and efficiency, making it suitable for practical applications. Compared to Dark Channel Prior, the method offers faster processing but may lack robustness in dense haze conditions [14].

He *et al*. (2010) introduced the Dark Channel Prior (DCP) method for single image haze removal. The method is based on the observation that in non-hazy images, at least one color channel contains pixels with very low intensity. Using this prior, the method estimates the transmission map and atmospheric light to restore clear images. The DCP method is simple, effective, and widely used in image dehazing. Experimental results show significant improvement in visibility and detail restoration. However, the method may produce halo artifacts near edges and requires refinement techniques such as guided filtering. Despite these limitations, DCP remains a fundamental technique and serves as the basis for many advanced dehazing methods [15].

Shuai *et al.* (2012) proposed an image haze removal method by combining Wiener filtering with the Dark Channel Prior (DCP). The main objective of this work is to improve the quality of dehazed images by reducing noise and enhancing clarity simultaneously. The DCP is used to estimate the transmission map and atmospheric light, while Wiener filtering is applied to suppress noise introduced during the restoration process. This combination improves both visibility and smoothness of the output image. Experimental results show that the proposed method produces better visual quality compared to basic DCP, especially in noisy environments. However, the approach increases computational complexity due to the additional filtering step. It may also require parameter tuning for optimal performance. Despite these limitations, this method enhances the effectiveness of DCP and demonstrates

the importance of integrating filtering techniques for improved dehazing results [16].

Pan *et al*. (2016) introduced a blind image deblurring method based on the Dark Channel Prior. The study focuses on restoring sharp images from blurred inputs caused by motion or atmospheric conditions. The authors observed that the dark channel property can also be applied to blurred images, where the intensity distribution differs from sharp images. By incorporating this prior into the deblurring framework, the method effectively estimates blur kernels and restores image details. Experimental results demonstrate significant improvement in sharpness and clarity compared to traditional deblurring techniques. However, the method is computationally intensive and may not perform well for extremely complex blur patterns. Although primarily designed for deblurring, the approach is relevant to image dehazing, as both problems involve restoration of degraded images. This work highlights the versatility of DCP and its application beyond haze removal [17].

Wang *et al*. (2023) proposed an optimized single image dehazing algorithm based on an improved Dark Channel Prior. The study aims to overcome common limitations of the traditional DCP method, such as halo artifacts and inaccurate transmission estimation. The authors introduced modifications to improve the estimation of atmospheric light and refine the transmission map. The enhanced algorithm produces better edge preservation and reduces visual artifacts. Experimental results show improved performance in terms of clarity, contrast, and color accuracy compared to the original DCP. However, the method involves additional computational steps, which may increase processing time. This work demonstrates that DCP can be further improved through optimization techniques. It provides valuable insights for developing advanced dehazing methods while maintaining the simplicity and effectiveness of the original approach [18].

Lee *et al*. (2016) presented a comprehensive review of image dehazing algorithms based on the Dark Channel Prior. The paper analyzes various improvements and modifications proposed to enhance the performance of DCP. The authors discussed key challenges such as halo artifacts, color distortion, and computational complexity. The review highlights different techniques used to overcome these issues, including guided filtering, soft matting, and optimization methods. The study also compares DCP with other dehazing approaches, providing a detailed understanding of its advantages and limitations. The paper concludes that DCP remains one of the most effective and widely used methods for single image dehazing. This review serves as a valuable reference for researchers and helps in

understanding the evolution of DCP-based techniques. It provides a strong theoretical background for developing improved dehazing algorithms [19].

Thirumala *et al*. (2021) proposed a haze-level prior approach to enhance object visibility under atmospheric degradation. The method focuses on estimating the level of haze present in an image and adjusting enhancement techniques accordingly. By incorporating haze-level information, the approach improves visibility and contrast more effectively than traditional methods. The model adapts to different atmospheric conditions, making it suitable for real-world applications. Experimental results show improved object visibility and detail preservation in degraded images. However, the method may require accurate estimation of haze levels, which can be challenging in complex scenes. The approach provides an alternative to traditional DCP-based methods by focusing on adaptive enhancement. This work highlights the importance of understanding haze characteristics for effective image restoration and can be used in combination with DCP for results [20].

Kumari *et al*. (2021) proposed a fast and efficient visibility restoration technique for single image dehazing and defogging. The method focuses on reducing computational complexity while maintaining high-quality output. The authors utilized an optimized transmission estimation process to recover clear images from degraded inputs. The approach improves contrast, visibility, and color consistency without requiring heavy computations. Experimental results show that the method performs well compared to traditional techniques in terms of both speed and visual quality. However, the method may face challenges in handling extremely dense haze conditions. This work emphasizes the importance of balancing efficiency and performance in real-time applications. It provides a practical alternative to complex dehazing models and can be compared with the Dark Channel Prior method for its simplicity and faster execution [21].

Zhang *et al.* (2026) introduced an infrared-assisted single-stage framework for joint restoration and fusion of visible and infrared images under hazy conditions. The method combines information from both visible and infrared images to improve restoration quality. Infrared images are less affected by haze, making them useful for enhancing degraded visible images. The proposed framework performs restoration and fusion simultaneously, resulting in improved clarity and detail preservation. Experimental results demonstrate superior performance compared to single-modality approaches. However, the requirement of both visible and infrared images limits its applicability in some scenarios. The method is

computationally complex and requires specialized sensors. Despite these limitations, the study highlights the benefits of multi-modal approaches in image restoration and provides insights for improving dehazing techniques beyond single image methods like DCP [22].

Liu and Wu (2026) proposed an intelligent hybrid machine learning framework for low-visibility reconstruction in airport transportation systems. The method combines multiple machine learning techniques to enhance images affected by fog and haze. The framework focuses on improving visibility for safety-critical applications such as airport navigation. Experimental results show significant improvements in clarity, contrast, and object detection accuracy. However, the method requires large datasets and computational resources for training and implementation. The approach demonstrates the effectiveness of hybrid learning models in handling complex degradation scenarios. Compared to traditional methods like Dark Channel Prior, this method offers better adaptability but involves higher complexity. This work highlights the growing importance of machine learning in real-world image restoration applications [23].

Sahu *et al*. (2022) presented a comprehensive survey on techniques for haze removal from degraded images. The paper reviews various methods, including traditional enhancement techniques, model-based approaches, and deep learning methods. The authors analyzed the strengths and limitations of each approach, focusing on performance, computational complexity, and applicability. The survey highlights that model-based methods like Dark Channel Prior are widely used due to their effectiveness and simplicity. However, deep learning methods provide better performance at the cost of higher computational requirements. The study also identifies key challenges such as halo artifacts, color distortion, and handling dense haze. This survey provides a detailed overview of the field and serves as a valuable reference for researchers. It helps in understanding the evolution of dehazing techniques and supports the development of improved methods [24].

Alsakar *et al*. (2025) presented a comprehensive review of underwater image restoration and enhancement techniques. The study discusses recent trends, challenges, and applications in underwater imaging. The authors analyzed various methods, including filtering techniques, model-based approaches, and deep learning models. Although focused on underwater images, the degradation characteristics are similar to atmospheric conditions such as fog and haze. The survey highlights challenges such as color distortion, low contrast, and scattering effects. The study emphasizes the importance of combining multiple techniques for effective restoration. It also discusses the limitations of existing methods and suggests future

research directions. This work provides useful insights for atmospheric image dehazing and can be related to methods like Dark Channel Prior [25].

Kim *et al*. (2019) proposed a fast single image dehazing method using saturation-based transmission map estimation. The approach estimates the transmission map based on saturation values, which reduces computational complexity. The method improves image clarity and contrast while maintaining color balance. Experimental results show that the proposed technique performs efficiently compared to traditional methods. However, the method may not be effective in scenes with low saturation variation. The approach provides a simple alternative to complex dehazing algorithms. Compared to Dark Channel Prior, it offers faster processing but may lack robustness in certain conditions. This work contributes to the development of efficient dehazing techniques [26].

Ehsan *et al*. (2021) proposed a single image dehazing technique using dual transmission maps and gradient-domain guided filtering. The method estimates two transmission maps to improve accuracy and combines them for better restoration. The guided filtering process helps in preserving edges and reducing artifacts. Experimental results show improved clarity, contrast, and detail preservation. However, the method involves additional computational steps, which may increase processing time. The approach provides better performance compared to single transmission map methods. This work demonstrates the importance of accurate transmission estimation in dehazing. It can be compared with Dark Channel Prior for performance evaluation [27].

Rahmawati *et al*. (2024) proposed a transmission map refinement technique using Laplacian transform based on the Dark Channel Prior method. The approach improves the accuracy of transmission estimation by refining it using Laplacian filtering. This helps in reducing halo artifacts and enhancing edge preservation. Experimental results show improved visual quality and detail restoration. However, the method increases computational complexity due to additional processing steps. The approach enhances the performance of traditional DCP and provides better results in challenging conditions. This work highlights the importance of refinement techniques in improving dehazing performance [28].

Debevec and Malik (2023) proposed a method for recovering high dynamic range (HDR) radiance maps from photographs. The approach combines multiple images taken at different exposure levels to create a single high-quality image. The method improves detail visibility in both dark and bright regions. Although not specifically designed for dehazing,

HDR techniques can enhance visibility in degraded images. The approach requires multiple images, which may limit its applicability. However, it provides valuable insights for improving image quality. This work contributes to the broader field of image enhancement [29].

Venkatesh *et al*. (2022) proposed an image enhancement technique using CLAHE (Contrast Limited Adaptive Histogram Equalization) and bilinear interpolation. The method improves contrast and enhances image details. CLAHE prevents over-amplification of noise while enhancing local contrast. Bilinear interpolation helps in smooth image scaling. Experimental results show improved visual quality compared to traditional methods. However, the method does not specifically address atmospheric degradation. It can be used as a preprocessing or post-processing step in dehazing systems. This work highlights the importance of contrast enhancement techniques in image processing [30].

CHAPTER 3

INTRODUCTION TO IMAGE PROCESSING

3.1 Introduction

In recent years, image processing has become an important area of research due to its wide range of applications in various fields such as surveillance, remote sensing, medical imaging, and computer vision. Images captured in outdoor environments are often affected by atmospheric conditions such as fog, smoke, and sand dust. These conditions degrade the quality of images by causing light scattering and absorption, which leads to reduced visibility, poor contrast, and color distortion. As a result, important details present in the scene are lost, making the images less useful for analysis and interpretation. To address these issues, image enhancement and restoration techniques are widely used to improve the visual quality of degraded images. These techniques aim to recover scene information and enhance visibility without introducing additional distortion. The process of restoring degraded images involves understanding the effects of atmospheric conditions and applying suitable models to compensate for these effects. This chapter presents an overview of image processing concepts related to image degradation and restoration. It also discusses the fundamental principles involved in improving image quality under adverse environmental conditions, which form the basis for further analysis and implementation in this work.

3.2 Image Formation and Processing

Image formation is the process by which a visual scene is captured and converted into a digital image. It begins with a light source, such as the sun or artificial lighting, illuminating objects in the environment. The light reflected from these objects passes through an optical system (lens) and is focused onto an image sensor, such as a CCD or CMOS sensor in a camera. The sensor converts the incoming light into electrical signals, where each pixel records the intensity of light at that location. These signals are then digitized into numerical values, forming a digital image as shown in the figure 3.1 . In color imaging, filters are used to separate light into Red, Green, and Blue (RGB) components, allowing the representation of color information. However, during image formation, various factors such as noise, poor lighting, and atmospheric conditions like fog, smoke, and sand dust can degrade the image quality. These

conditions cause scattering and absorption of light, leading to reduced contrast, blurred details, and color distortion. As a result, the captured image may not accurately represent the original scene.

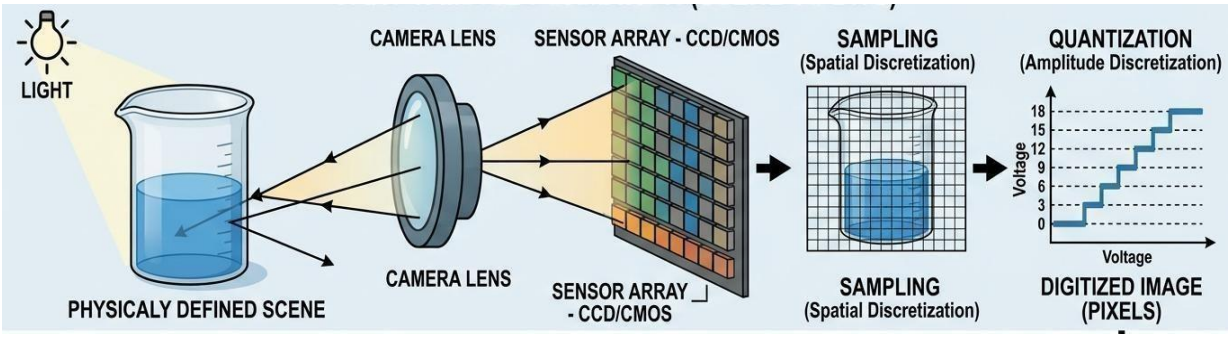

Figure 3.1: Image Formation

Image processing is the technique used to improve the quality of such degraded images and to extract useful information. It involves a series of operations applied to the digital image. The first step is image acquisition, where the image is obtained from a camera or dataset. This is followed by preprocessing, which includes noise reduction, resizing, and normalization to prepare the image for further processing. Next, enhancement techniques are applied to improve the visual quality of the image. These include contrast enhancement, brightness adjustment, and sharpening. In the case of degraded images affected by fog, smoke, or dust, restoration techniques are used to recover the original scene information by compensating for atmospheric effects. Further processing may include segmentation, where the image is divided into meaningful regions, and feature extraction, where important characteristics such as edges, textures, or shapes are identified. These steps are useful for higher-level tasks such as object detection and recognition.

Finally, the processed image is evaluated using quality metrics to determine the effectiveness of the applied techniques. Thus, image processing plays a crucial role in transforming raw images into clear and meaningful visual information for various real-world applications.

3.3 Types of Images in Image Processing

In image processing, images are classified into different types based on their pixel representation, color information, and application. Understanding these image types is essential

for selecting appropriate enhancement and restoration techniques.

3.3.1 Binary Image

A binary image is a type of digital image that contains only two possible pixel values, typically 0 and 1, representing black and white respectively. These images are generated by applying a thresholding technique on grayscale or color images, where pixels are classified into foreground and background as shown in the figure 3.2. Binary images are widely used in image processing applications such as object detection, segmentation, and pattern recognition due to their simplicity and reduced computational complexity. They are especially useful in extracting shapes and structures from images.

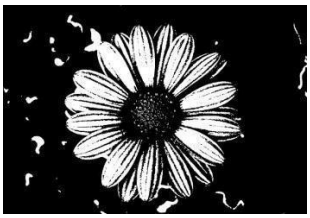

Figure 3.2: Binary Image

3.3.2 Gray Scale image

A grayscale image is a type of digital image that contains only intensity information without any color like shown below figure 3.3. Each pixel in a grayscale image represents a brightness level ranging from 0 to 255, where 0 corresponds to black and 255 corresponds to white. These images are simpler compared to color images and require less computational complexity, making them widely used in image processing applications such as edge detection, feature extraction, and medical imaging. Grayscale images help in focusing on structural details rather than color information, which is useful in analysis tasks.

Figure 3.3: Grayscale Image

3.3.3 Colour Image (RGB Image)

An RGB image is a type of color image that is formed by combining three primary color channels: Red (R), Green (G), and Blue (B) as shown in figure 3.4. Each pixel in an RGB image consists of three intensity values corresponding to these channels, typically ranging from 0 to 255. By varying the combination of these values, a wide range of colors can be represented. RGB images are widely used in digital cameras, computer displays, and image processing applications. They provide detailed color information, making them suitable for tasks such as object recognition and visual analysis.

Figure 3.4: Colour Image

3.3.4 HSV Images

HSV (Hue, Saturation, Value) is a color representation model used in image processing that describes colors in a way that is closer to human perception compared to the RGB model. Instead of representing colors using Red, Green, and Blue components, the HSV model separates color information into three distinct components: Hue, Saturation, and Value as shown in figure 3.5.

- **Hue (H)** represents the type of color and is measured in degrees ranging from 0° to 360°. For example, 0° corresponds to red, 120° to green, and 240° to blue.
- **Saturation (S)** represents the intensity or purity of the color. It ranges from 0 to 1, where 0 indicates a completely desaturated (gray) color and 1 indicates a fully saturated color.
- **Value (V)** represents the brightness or intensity of the color, ranging from 0 (black) to 1 (maximum brightness).

Figure 3.5: HSV Image

The HSV color model is widely used in image enhancement, segmentation, and object detection because it separates color and intensity information. This makes it easier to process images affected by lighting variations, fog, smoke, or dust.

3.3.5 Multispectral Image

A multispectral image is a type of image that captures data at multiple wavelengths across the electromagnetic spectrum, beyond the visible RGB channels. These images typically consist of several bands, such as visible, infrared, and sometimes ultraviolet, allowing detailed analysis of objects and materials as shown in figure 3.6. Each band provides specific information that is not visible to the human eye. Multispectral imaging is widely used in applications such as remote sensing, agriculture, environmental monitoring, and satellite imaging. It helps in identifying features like vegetation health, water bodies, and land cover.

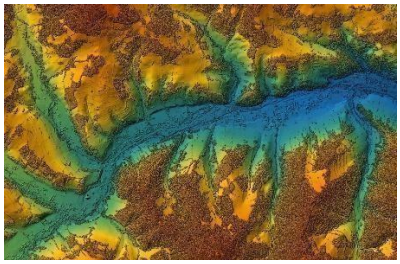

Figure 3.6: Multispectral Image

In the image processing different types of images can be characterized based on their channels, pixel value ranges, and data representation. A binary image consists of a single channel with pixel values of either 0 (black) or 1 (white), and it is represented using 1-bit data. A grayscale image also has one channel, but its pixel values range from 0 to 255, typically represented using 8 bits. An RGB image contains three channels—red, green, and blue—where each channel has pixel values ranging from 0 to 255, resulting in a 24-bit representation. Similarly, an HSV image also has three channels, where hue ranges from 0 to 360 degrees, and saturation and value range from 0 to 1, forming a cylindrical coordinate representation. In contrast, multispectral images

consist of multiple channels, usually ranging from 3 to more than 15 bands, and their pixel values represent radiance or reflectance, making them high-dimensional in nature.

Image Type	Channel	Pixel Value Range	Data Representation
Binary	1	0 (Black) or 1 (White)	1-bit
Grayscale	1	0 to 255	8-bits
RGB	3	0 - 255 (per channel)	24-bits
HSV	3	H: 0-360°, S/V:0-1	Cylindrical Coordinate
Multispectral	3 - 15+	Radiance / Reflectance	High-Dimensional

Table I: Comparsion of Different types of Images

3.4 Image Quality Metrics

In image processing, the performance of enhancement and restoration techniques is evaluated using various quantitative metrics. These metrics help in measuring the quality of the processed image by comparing it with the original or reference image. The commonly used image quality metrics include Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index Measure (SSIM).

3.4.1 Mean Squared Error (MSE)

Mean Squared Error (MSE) is one of the most basic and widely used metrics for evaluating the quality of an image after processing. It measures the average squared difference between the pixel values of the original image and the processed (enhanced or restored) image. MSE helps in quantifying the amount of error introduced during image processing.

The MSE is defined as:

$$MSE = \frac{1}{MN} \sum_{i=1}^{M} \sum_{j=1}^{N} [I(i,j) - K(i,j)]^2$$

Where:

- $I(i,j)$is the original image pixel value
- $K(i,j)$is the processed (or reconstructed) image pixel value
- $M \times N$is the image size

A lower MSE value indicates that the processed image is very close to the original image, meaning better quality and less distortion. On the other hand, a higher MSE value indicates more error and lower image quality.

3.4.2 Peak Signal-to-Noise Ratio (PSNR)

Peak Signal-to-Noise Ratio (PSNR) is one of the most widely used metrics for evaluating the quality of an enhanced or restored image. It measures the ratio between the maximum possible power of a signal (original image) and the power of noise (error introduced during processing). PSNR is expressed in decibels (dB) and is used to assess how closely the processed image resembles the original image.

The PSNR is calculated using the Mean Squared Error (MSE) between the original and processed images. It is defined as:

$$PSNR = 10\log_{10}\left(\frac{MAX^2}{MSE}\right)$$

where MAX represents the maximum possible pixel value of the image (for an 8-bit image,$MAX = 255$), and MSE is the mean squared error between the original and processed images. A higher PSNR value indicates better image quality, meaning the processed image is closer to the original image with less distortion. Typically, a PSNR value above 30 dB is considered acceptable, while values above 40 dB indicate high-quality reconstruction.

3.4.3 Structural Similarity Index Measure (SSIM)

Structural Similarity Index Measure (SSIM) is an advanced image quality assessment metric used to evaluate the similarity between an original image and a processed image. Unlike traditional metrics such as MSE and PSNR, SSIM considers the perceptual quality of the image

by taking into account structural information, luminance, and contrast. This makes it more consistent with human visual perception.

SSIM compares local patterns of pixel intensities that have been normalized for luminance and contrast. It is defined as:

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Where:

- μ_x, μ_y are the mean intensities of images x and y
- σ_x^2, σ_y^2 are the variances
- σ_{xy} is the covariance
- C_1, C_2 are constants to stabilize the division

The SSIM value ranges between 0 and 1, where a value closer to 1 indicates high similarity between the images. A higher SSIM value means that the processed image preserves structural information effectively. SSIM is widely used in image processing applications such as dehazing, denoising, and compression, as it provides a more accurate assessment of visual quality compared to traditional error-based metrics.

CHAPTER 4

MATLAB SOFTWARE

4.1 Introduction

For my project we focuses on the implementation, processing, and analysis of image enhancement techniques using MATLAB (Matrix Laboratory) software. MATLAB is a powerful numerical computing environment widely used for image processing, algorithm development, and visualization. It provides a user-friendly platform for handling complex mathematical operations and processing digital images efficiently.

The MATLAB environment includes a specialized Image Processing Toolbox, which offers a wide range of built-in functions for reading, analyzing, enhancing, and displaying images. These tools enable effective processing of images degraded by atmospheric conditions such as fog, smoke, and sand dust. Using MATLAB, various image enhancement operations such as filtering, contrast adjustment, and intensity transformation can be performed with high accuracy and flexibility.

The complete workflow involves:

- Importing the input image into the MATLAB workspace.
- Performing image preprocessing operations.
- Converting the image into required color spaces.
- Applying filtering techniques for noise removal.
- Estimating degradation parameters.
- Performing image enhancement operations.
- Applying post-processing techniques.
- Evaluating performance using metrics.
- Visualizing input and output images.
- Saving the processed image results.

MATLAB supports matrix-based operations, which are highly suitable for image representation and manipulation. It enables efficient implementation of algorithms and simplifies the processing of large datasets. Due to its accuracy, ease of use, and extensive library support, MATLAB is widely preferred by researchers and engineers in the field of image processing.

4.2 MATLAB Software Overview

MATLAB is a powerful numerical computing and programming environment widely used for image processing, data analysis, and algorithm development. With its advanced Image Processing Toolbox, it provides efficient tools for analyzing, enhancing, and restoring digital images. MATLAB supports matrix-based computations, enabling accurate and fast implementation of image processing techniques. It is widely used by researchers and engineers for developing and testing algorithms related to image enhancement, including applications involving fog, smoke, and sand dust degraded images.

Figure 4.1: MATLAB software Logo

4.2.1 Key Capabilities of MATLAB

MATLAB performs advanced digital image processing by applying mathematical and algorithmic operations on image data to enhance visual quality and extract meaningful information. Using its powerful Image Processing Toolbox, it enables tasks such as image enhancement, filtering, segmentation, and feature extraction with high accuracy. The software processes images as matrices, allowing efficient manipulation and transformation of pixel values. This ensures precise improvement of image parameters such as contrast, sharpness, and visibility. This ensures precise improvement of image parameters such as contrast, sharpness, and visibility. MATLAB also supports quantitative evaluation using metrics like PSNR, SSIM, and MSE, which helps in validating the quality of processed images.

- **Image Acquisition**

Image acquisition is the initial step in MATLAB-based image processing, where input images are collected and imported into the workspace. MATLAB supports various formats such as JPEG, PNG, BMP, and TIFF using functions like *imread*. In this project, degraded images affected by fog, smoke, and sand dust are taken as input. Proper acquisition ensures that the image is correctly loaded without any loss of information. This step is important because the quality and format of the input image directly influence the effectiveness of subsequent processing stages. Accurate acquisition provides a reliable foundation for enhancement and restoration processes.

- **Preprocessing**

Preprocessing involves preparing the acquired image for further analysis and enhancement. In MATLAB, this includes resizing the image, converting color spaces (such as RGB to grayscale or HSV), and reducing noise using filters. These operations help standardize the image and remove unwanted distortions that may affect processing accuracy. Preprocessing improves image quality and makes important features more distinguishable. It also reduces computational complexity by simplifying the image data. This step is essential because it ensures that the image is in an optimal form for applying enhancement techniques, leading to better and more reliable results.

- **Image Enhancement**

Image enhancement focuses on improving the visual quality of degraded images. MATLAB provides various techniques such as contrast adjustment, histogram equalization, and filtering methods to enhance visibility. In dehazing applications, this step plays a crucial role in restoring clarity by reducing the effects of fog, smoke, and sand dust. Enhancement improves brightness, contrast, and sharpness while preserving important details. The goal is to make the image more suitable for human interpretation and further analysis. Effective enhancement significantly improves the overall appearance of the image and ensures that important features are clearly visible.

- **Processing and Analysis**

In this step, advanced image processing techniques are applied to analyze and extract meaningful information from the image. MATLAB supports operations such as filtering, transformation, segmentation, and feature extraction. These techniques help in identifying important structures and improving image details. Processing enhances the effectiveness of the image by refining its quality and structure. Analysis allows comparison between original and processed images to understand improvements. This step is important for achieving accurate results and ensuring that the applied techniques work effectively under different conditions such as fog, smoke, and dust.

- **Performance Evaluation**

Performance evaluation is the final step where the quality of the processed image is assessed using quantitative metrics. MATLAB provides tools to calculate metrics such as PSNR (Peak Signal-to-Noise Ratio), SSIM (Structural Similarity Index), and MSE (Mean Square Error). These metrics help measure the level of improvement and similarity between the original and enhanced images. Higher PSNR and SSIM values indicate better quality, while lower MSE indicates fewer errors. This step is essential for validating the effectiveness of the image processing techniques. It ensures that the enhancement process produces accurate and reliable results suitable for practical applications.

4.3 Running the Simulation

Running code in MATLAB is a simple process. First, open MATLAB and set the working directory to the folder containing your script and input files. Then, open the script file (such as a .m file) in the Editor and check for any errors. Click the Run button to execute the code. During execution, results appear in the Command Window and Figure Window. If any errors occur, correct them and run the script again. Finally, verify the output and save the processed results for future use and documentation.

4.4 Extracting Image Processing Parameters

After running the MATLAB simulation, the performance of the image enhancement algorithm is evaluated using various quality metrics. These parameters help in analyzing the effectiveness of the dehazing process in terms of clarity, contrast improvement, detail preservation, and overall visual quality. MATLAB provides tools to generate visual outputs and quantitative measures for proper evaluation. The following sections describe the extraction and analysis of important performance metrics such as PSNR, SSIM, MSE, histogram analysis, contrast improvement, and visual comparison.

4.4.1 MSE (Mean Squared Error) Evaluation

MSE measures the average squared difference between the original and enhanced images. It indicates the amount of error introduced during processing.

A lower MSE value represents better image quality and fewer errors.

To evaluate MSE in MATLAB, follow these steps:

- Compute the error between original and processed images.
- Analyze the numerical value obtained.
- Lower values indicate better performance.

For calculated the MSE value we used,

```
mse_value = immse(original_image, processed_image);
```

MSE is often used along with PSNR to provide a complete understanding of image quality.

4.4.2 PSNR (Peak Signal-to-Noise Ratio)

PSNR is an important parameter used to measure the quality of the enhanced image compared to the original image. It indicates how much noise or distortion is present after processing. A higher PSNR value represents better image quality with minimal distortion. To analyze PSNR in MATLAB, follow these steps:

- Compute the PSNR value between the original and enhanced image.
- Observe the numerical value obtained for quality evaluation.
- Higher PSNR values indicate better enhancement performance.

For calculated the PSNR value we used,

```
psnr_value = psnr(processed_image, original_image);
```

Generally, a PSNR value above 30 dB is considered acceptable for good quality images. Higher values indicate that the enhancement process has preserved important image details while reducing degradation effects.

4.4.3 SSIM (Structural Similarity Index Measure)

SSIM is used to measure the similarity between the original and processed images based on structure, brightness, and contrast. Unlike PSNR, SSIM focuses more on human visual perception. For effective enhancement, SSIM values should be close to 1, which indicates high similarity and better structural preservation.

To analyze SSIM in MATLAB, follow these steps:

- Calculate the SSIM value between input and output images.
- Observe the similarity index obtained.
- Ensure the value is closer to 1 for better results.

For calculated the SSIM value we used,

```
ssim_value = ssim(processed_image, original_image);
```

If the SSIM value is low, it indicates loss of structural information, and further improvement in the enhancement algorithm may be required.

CHAPTER 5

PROPOSED METHODOLOGY

5.1 Introduction

Here we explain the proposed work developed for enhancing images degraded by atmospheric conditions such as fog, smoke, and sand dust. These conditions significantly reduce image quality by causing poor visibility, low contrast, color distortion, and loss of important details. Such degradation affects both human perception and the performance of computer vision systems. Therefore, it is essential to apply effective image processing techniques to restore clarity and improve overall image quality. The proposed methodology focuses on applying and analyzing three different image enhancement techniques: Non-Local Means (NLM) filtering, k-Singular Value Decomposition (k-SVD) combined with Contrast Limited Adaptive Histogram Equalization (CLAHE), and Proposed Method (Dark Channel Prior (DCP) combined with CLAHE). Each of these methods addresses specific challenges associated with degraded images and contributes to improving visual quality. By combining these approaches, the proposed work aims to achieve improved visibility, enhanced contrast, and better detail preservation. The performance of these methods is evaluated using standard metrics such as PSNR, SSIM, and MSE, along with visual comparison, to ensure accurate and reliable image enhancement.

5.2 Non-Local Means (NLM) filtering

The Non-Local Means (NLM) filter is an advanced image denoising technique widely used in image processing to remove noise while preserving important details and textures. Unlike traditional filters that rely only on local neighborhood information, the NLM filter considers the similarity between image patches across the entire image. This unique property makes it highly effective for restoring degraded images affected by noise, fog, smoke, and sand dust.

5.2.1 Mathematical Representation

NLM mathematical equation is given by:

$$NL[v](i) = \sum_{j \in I} w(i,j) \cdot v(j)$$

Where:

- $NL[v]$→Output filtered value of pixel i (new denoised pixel value)
- $v(j)$ → Intensity/value of neighboring pixel j
- $w(i,j)$→Weight between pixel i and pixel j (shows how similar the two pixels are)
- $\sum$ → Summation (adds contributions from all selected pixels)
- $j \in I$ → Pixel j belongs to image/search window I
- i → Current pixel being processed
- I → Search area or entire image region considered

5.2.2 Working Principle of NLM Filter

The fundamental concept behind the Non-Local Means (NLM) filter is based on the observation that natural images contain repetitive patterns and structures. Instead of relying only on nearby pixels for denoising, the NLM filter searches for similar pixel neighborhoods, known as patches, across the entire image. For each pixel, a small patch surrounding it is selected and compared with patches centered at other pixels in the image. The similarity between these patches is computed using a distance measure, typically the Euclidean distance. Based on this similarity, weights are assigned such that patches with higher similarity contribute more to the final pixel value, while dissimilar patches contribute less. The denoised pixel value is then obtained as a weighted average of all pixels. This non-local strategy allows the filter to effectively reduce noise while preserving important image structures such as edges and textures. In addition to denoising, image enhancement can be further improved using Contrast Limited Adaptive Histogram Equalization (CLAHE). CLAHE works by dividing the image into small regions called tiles and applying histogram equalization to each tile independently, thereby enhancing local contrast. To prevent over-amplification of noise, CLAHE limits the contrast by clipping the histogram at a predefined threshold before redistribution. When combined with NLM filtering, CLAHE enhances the visibility of fine details and textures in the denoised image, resulting in improved overall image quality with better contrast and reduced noise.

5.2.3 Software Description

```
clc;
clear;
close all;
img = imread('027.png');  % Replace with y
our image img = im2double(img);
lab = rgb2lab(img);
L = lab(:,:,1) / 100;   % Normalize L (0–1)
A = lab(:,:,2);
B = lab(:,:,3);
L_denoised = imnlmfilt(L, 'DegreeOfSmoothing', 0.01);
L_clahe = adapthisteq(L_denoised, ...
                'NumTiles',[8 8], ...
                'ClipLimit',0.02);
L_final = imadjust(L_clahe, stretchlim(L_clahe,[0.01 0.99]), []);
lab_enhanced(:,:,1) = L_final * 100;

lab_enhanced(:,:,2) = A;
lab_enhanced(:,:,3) = B;
final_img = lab2rgb(lab_enhanced);
figure;
subplot(1,2,1);
imshow(img);
title('Original Low-Light Image');

subplot(1,2,2);
imshow(final_img);
title('Enhanced Image (NLM + CLAHE)');
```

5.2.4 Overview of NLM Filter

The Non-Local Means (NLM) filter is an effective image denoising technique that preserves fine details and textures by utilizing similarities between image patches across the

entire image. It reduces noise while maintaining edge sharpness and natural appearance, making it suitable for enhancing images affected by fog, smoke, and sand dust. However, it has limitations such as high computational complexity, sensitivity to parameter selection, and increased memory requirements. Despite these challenges, optimized versions improve its efficiency. The NLM filter is widely used in image denoising, medical imaging, remote sensing, and computer vision, and in this project, it plays a key role in improving image clarity while preserving important details.

5.3 k-SVD combined with CLAHE

The k-SVD (K-Singular Value Decomposition) algorithm is an advanced image denoising technique widely used in image processing to remove noise while preserving important structures and details. Unlike traditional filtering methods, k-SVD is based on sparse representation, where image patches are expressed as a combination of a few basis elements from a learned dictionary. This adaptive dictionary learning capability allows the algorithm to effectively distinguish between noise and meaningful image content. As a result, k-SVD is highly effective in restoring degraded images affected by noise, fog, smoke, and sand dust while maintaining sharp edges and fine textures.

5.3.1 Mathematical Representation

The k-SVD optimization problem is defined as:

$$\min_{D,\alpha} \| Y - D\alpha \|_2^2 \text{ subject to } \| \alpha_i \|_0 \leq T_0$$

The CLAHE transformation is based on histogram equalization:

$$s = (L-1) \int p_r(w)\, dw$$

where:

- Y represents image patches,
- D is the learned dictionary,
- α is the sparse coefficient,
- T_0 is the sparsity constraint,
- r and $s$are input and output pixel intensities,
- L is the number of gray levels.

5.3.2 Working Principle of k-SVD with CLAHE

The combined method operates in two main stages:

1. **k-SVD Denoising Stage**

   The input image is first divided into overlapping patches. Each patch is represented as a sparse combination of dictionary atoms. The k-SVD algorithm iteratively updates the dictionary and sparse coefficients to remove noise while preserving important structures.

2. **CLAHE Enhancement Stage**

   After denoising, the output image is processed using CLAHE. The image is divided into small tiles, and histogram equalization is applied to each tile. A contrast limiting factor is used to prevent noise amplification. Finally, interpolation is performed to ensure smooth transitions between tiles.

   This two-stage process ensures effective noise removal followed by contrast enhancement, resulting in high-quality images.

5.3.3 Software Description

```
clear; close all; clc;
img = imread('024.png');
if size(img,3)==3
    img = rgb2gray(img);
end
img = im2double(imresize(img,[256 256]));
figure; imshow(img); title('Original');
patchSize = 8;
step = 1;      % sliding step
K = 4;         % sparsity level
D = kron(dctmtx(patchSize), dctmtx(patchSize));
D = D ./ vecnorm(D);
[h,w] = size(img);
patches = [];
positions = [];
for i = 1:step:h-patchSize+1
```

```
    for j = 1:step:w-patchSize+1
        patch = img(i:i+patchSize-1, j:j+patchSize-1);
        patches(:,end+1) = patch(:);
        positions(:,end+1) = [i; j];
    end
end
numPatches = size(patches,2);
coeffs = zeros(size(D,2), numPatches);
for i = 1:numPatches
    coeffs(:,i) = omp(D, patches(:,i), K);
end
reconPatches = D * coeffs;
reconstructed = zeros(h,w);
weight = zeros(h,w);
for k = 1:numPatches
    i = positions(1,k);
    j = positions(2,k);
    block = reshape(reconPatches(:,k), patchSize, patchSize);
    reconstructed(i:i+patchSize-1, j:j+patchSize-1) = ...
        reconstructed(i:i+patchSize-1, j:j+patchSize-1) + block;
    weight(i:i+patchSize-1, j:j+patchSize-1) = ...
        weight(i:i+patchSize-1, j:j+patchSize-1) + 1;
end
reconstructed = reconstructed ./ weight;
figure;
subplot(1,2,1); imshow(img); title('Original');
subplot(1,2,2); imshow(reconstructed);
title('Sparse Reconstruction');
figure; imshow(abs(img-reconstructed),[]);
title('Difference');
L_enhanced = adapthisteq(reconstructed, ...
    'ClipLimit', 0.5, ...
```

```
    'Distribution', 'rayleigh');
figure; imshow(abs(L_enhanced),[]);
title('Enhanced output');
function x = omp(D, y, sparsity)
residual = y;
indx = [];
x = zeros(size(D,2),1);
for k = 1:sparsity
    proj = D' * residual;
    [~, pos] = max(abs(proj));
    indx = unique([indx pos]);
    subDict = D(:,indx);
    a = pinv(subDict) * y;
    residual = y - subDict*a;
    if norm(residual) < 1e-6
        break;
    end
end
x(indx) = a;
end
```

5.3.4 Overview of K-SVD with CLAHE

The combination of K-SVD and CLAHE is a powerful hybrid image enhancement technique that integrates denoising and contrast enhancement into a single framework. K-SVD effectively removes noise while preserving important structural details and textures, whereas CLAHE enhances local contrast and improves visibility. This approach is particularly useful for images degraded by fog, smoke, and sand dust, where both clarity and contrast are essential. Although the method involves higher computational complexity and requires careful parameter tuning, it produces superior visual results. In this project, K-SVD with CLAHE plays a significant role in enhancing image quality by improving both detail preservation and contrast.

5.4 Dark Channel Prior (DCP) combined with CLAHE

The Dark Channel Prior (DCP) with CLAHE is a powerful image enhancement technique widely used for removing haze and improving visibility in degraded images. It is especially effective for images affected by atmospheric conditions such as fog, smoke, and sand dust. Unlike traditional enhancement methods, DCP is based on a physical model of light propagation in hazy environments. It estimates the effect of atmospheric scattering and restores the original scene radiance. When combined with Contrast Limited Adaptive Histogram Equalization (CLAHE), the method further enhances local contrast and improves the visual quality of the image.

5.4.1 Mathematical Representation

The DCP method is based on the atmospheric scattering model, which is expressed as:

$$I(x) = J(x)\, t(x) + A\,(1 - t(x))$$

where:

- $I(x)$is the observed hazy image,
- $J(x)$is the scene radiance (desired clear image),
- $t(x)$is the transmission map,
- $A$is the global atmospheric light.

5.4.2 Working Principle of DCP with CLAHE

The working principle of the DCP method begins with the computation of the dark channel of the input image. This involves identifying the minimum intensity value among the color channels in a local patch. Using this information, the algorithm estimates the transmission map, which represents the portion of light that reaches the camera without being scattered. The atmospheric light is also estimated, which corresponds to the brightest pixels in the hazy image. Once the transmission map and atmospheric light are determined, the scene radiance is recovered using an image formation model. This process effectively removes haze and restores the original colors and details of the image. After dehazing, CLAHE is applied to further enhance the image. CLAHE works by dividing the image into small regions and applying histogram equalization to each region while limiting contrast amplification to avoid noise enhancement. This combination ensures both haze removal and contrast improvement,

resulting in a clear and visually appealing image.

5.4.3 Software Description

```
clc;
clear;
close all;
I = imread('baz.jpg');
I = im2double(I);
[m,n,~] = size(I);
figure subplot(2,3,1)
imshow(I)
title('Original Hazy Image')
patch = 15;
dark = zeros(m,n);

for i = 1:m
    for j = 1:n
        rmin = max(i-floor(patch/2),1);
        rmax = min(i+floor(patch/2),m);
        cmin = max(j-floor(patch/2),1);
        cmax = min(j+floor(patch/2),n);
        patchImg = I(rmin:rmax , cmin:cmax , :);
        dark(i,j) = min(patchImg(:));
    end
end
subplot(2,3,2)
imshow(dark)
title('Dark Channel')
darkVec = dark(:);
imageVec = reshape(I,m*n,3);
[~,index] = sort(darkVec,'descend');
```

```
  numPixels = floor(m*n*0.001);
  A = mean(imageVec(index(1:numPixels),:));
  omega = 0.95;
  transmission = zeros(m,n);
  for i = 1:m
     for j = 1:n
        transmission(i,j) = 1 - omega*dark(i,j)/max(A);
     end

        end

subplot(2,3,3)
  imshow(transmission)
  title('Initial Transmission')
  kernel = fspecial('average',[15 15]);
  t_refined = imfilter(transmission,kernel,'replicate');
  subplot(2,3,4)
  imshow(t_refined)
  title('Refined Transmission')
  t0 = 0.01;
  J = zeros(size(I));
  for c = 1:3
     J(:,:,c) = (I(:,:,c) - A(c)) ./ max(t_refined,t0) + A(c);
  end
  J(J>1) = 1;
  J(J<0) = 0;
  J = imadjust(J,stretchlim(J),[]);
  subplot(2,3,5)
  imshow(J)
  title('Dehazed Image')
  R = J(:,:,1);
  G = J(:,:,2);
  B = J(:,:,3);
```

```
[countR,binR] = imhist(R);
[countG,binG] = imhist(G);
[countB,binB] = imhist(B);
subplot(2,3,6)
plot(binR,countR,'r','LineWidth',1.5)
hold on
plot(binG,countG,'g','LineWidth',1.5)
plot(binB,countB,'b','LineWidth',1.5)
xlabel('Bins')
ylabel('# of Pixels')
title('Dehazed RGB Histogram')
legend('Red','Green','Blue')
figure
imshowpair(I,J,'montage')
title('Before and After Dehazing')
```

5.4.4 Overview of DCP with CLAHE

The combination of Dark Channel Prior (DCP) with CLAHE is an effective image enhancement technique that focuses on dehazing and contrast improvement. DCP restores image clarity by estimating and removing atmospheric effects, while CLAHE enhances local contrast and reveals hidden details. This method preserves edges and fine structures, making it suitable for images affected by fog, smoke, and dust. Although it may face challenges such as inaccurate estimation in bright regions, higher computational cost, and sensitivity to parameter tuning, it still provides significant improvement in visual quality. In this project, DCP with CLAHE plays an important role in enhancing visibility and producing clearer, more informative images.

CHAPTER 6
RESULTS AND DISCUSSIONS

In this results and discussion session, foggy, smoke and sand-dust degraded images are processed to obtain clear and enchanced images. We comparing the enhanced images with the corresponding ground truth images obtained from the Kaggle dataset. The ground truth images represent clear, haze-free reference images and serve as a benchmark for evaluating the effectiveness of the enhancement methods. In this work, two different degraded input images affected by fog, smoke, and sand dust are considered to validate the performance of the implemented methods.

The presented figures illustrate a detailed comparison of multiple image enhancement and restoration techniques applied to degraded input images, with corresponding ground truth images serving as references for ideal output quality. These results highlight how different algorithms address common image degradation issues such as noise, low contrast, and haze, and how effectively they restore visual clarity and structural details.

Figures 6.1.1 and 6.2.1 represent the original input images, which are visibly affected by degradation. These degradations may include noise, poor illumination, low dynamic range, and atmospheric effects like haze or fog. As a result, important features in the images are not clearly distinguishable. In contrast, Figures 6.1.2 and 6.2.2 show the ground truth images, which act as benchmarks for evaluating performance. These images exhibit optimal clarity, enhanced contrast, and well-preserved details, making them suitable references for assessing how closely each method approximates the desired output.

The first enhancement approach applied is Non-Local Means (NLM) combined with Contrast Limited Adaptive Histogram Equalization (CLAHE), as shown in Figures 6.1.3 and 6.2.3. The NLM algorithm is a denoising technique that works by averaging similar pixel neighborhoods across the entire image rather than relying solely on local information. This allows it to effectively reduce noise while preserving repetitive structures and textures. Unlike traditional filters, NLM considers the similarity between patches, making it particularly effective for images with structured patterns. After denoising, CLAHE is applied to improve contrast. CLAHE enhances local contrast by redistributing pixel intensity values within small regions, while also limiting amplification to avoid excessive noise enhancement. The combination of

NLM and CLAHE produces images with significantly reduced noise and improved brightness. However, one limitation observed in the figures is that fine details may appear slightly blurred due to the averaging process in NLM. While the overall visual quality is improved compared to the input, some sharp edges and intricate textures are not fully preserved, indicating a compromise between noise removal and detail retention.

The second method evaluated is k-SVD combined with CLAHE, as shown in Figures 6.1.4 and 6.2.4. The k-SVD algorithm is a dictionary learning-based approach used for sparse representation of image patches. It works by decomposing the image into small patches and representing each patch as a sparse combination of basis elements (atoms) from a learned dictionary. This adaptive dictionary is trained iteratively to best represent the image content, enabling effective noise removal while preserving important structures. Compared to NLM, k-SVD provides a more sophisticated approach to denoising, as it captures underlying patterns and textures in a more flexible and data-driven manner. After denoising, CLAHE is again applied to enhance contrast. The results demonstrate that k-SVD with CLAHE achieves better preservation of edges and fine details compared to NLM. The images appear sharper, with improved structural clarity and reduced noise. However, minor artifacts may still be present in certain regions, especially where the sparse representation does not perfectly model the image content. Additionally, k-SVD is computationally more complex than NLM, which may limit its practicality in real-time applications. Despite these limitations, the method shows a strong balance between denoising and detail preservation.

The third technique presented is Dark Channel Prior (DCP) combined with CLAHE, as shown in Figures 6.1.5 and 6.2.5. The DCP method is primarily designed for image dehazing and is based on the observation that in most natural haze-free images, at least one color channel in a local patch has very low intensity values. Using this prior, the algorithm estimates the thickness of haze in different regions of the image and reconstructs a haze-free version by recovering the scene radiance. This method is particularly effective in improving visibility in images affected by fog, mist, or atmospheric scattering. Once the dehazing process is complete, CLAHE is applied to further enhance contrast and bring out local details. The results clearly show that DCP with CLAHE produces the most visually striking improvements among the three methods. The images exhibit significantly enhanced contrast, improved depth perception, and clearer visibility of distant objects. Edges and textures are more pronounced, making the output

closer to the ground truth images. However, the method is not without limitations. In some cases, DCP may introduce halo artifacts around strong edges or lead to slight over-enhancement, especially in regions with dense haze or uneven illumination. Despite these issues, the overall performance of DCP with CLAHE is superior in terms of restoring visibility and contrast.

Figure 6.1.1: Input Image

Figure 6.1.2: Ground Truth Image

Figure 6.1.3: NLM with CLAHE

Figure 6.1.4: k-SVD with CLAHE

Figure 6.1.5: DCP with CLAHE

Figure 6.2.1: Input Image

Figure 6.2.2: Ground Truth Image

Figure 6.2.3: NLM with CLAHE

Figure 6.2.4: k-SVD with CLAHE

Figure 6.2.5: DCP with CLAHE

Figures 6.3 and 6.4 present the performance of the proposed enhancement technique on real-time captured images, demonstrating its practical applicability beyond standard benchmark datasets. Figure 6.3 shows the original real-time noisy image acquired from a live environment. The image contains visible noise, reduced contrast, and loss of fine details caused by environmental conditions, sensor limitations, and illumination variations during image acquisition. Such degradations are commonly observed in real-time imaging systems, where

uncontrolled lighting and motion can significantly affect image quality. As a result, important visual features in the original image are partially obscured, making interpretation and analysis more difficult.

Figure 6.4 illustrates the output produced by the proposed method when applied to the same real-time noisy image. The enhanced image shows a substantial reduction in noise while preserving essential structural information. The proposed algorithm effectively improves local contrast, making hidden details more visible and restoring natural brightness across the scene. Compared to the original image, edges appear sharper and textures become more distinguishable, which indicates that the method successfully suppresses unwanted distortions without excessive smoothing. Another important observation is that the enhanced image maintains a balanced visual appearance, avoiding over-amplification of brightness or artificial-looking artifacts that can occur in some enhancement techniques.

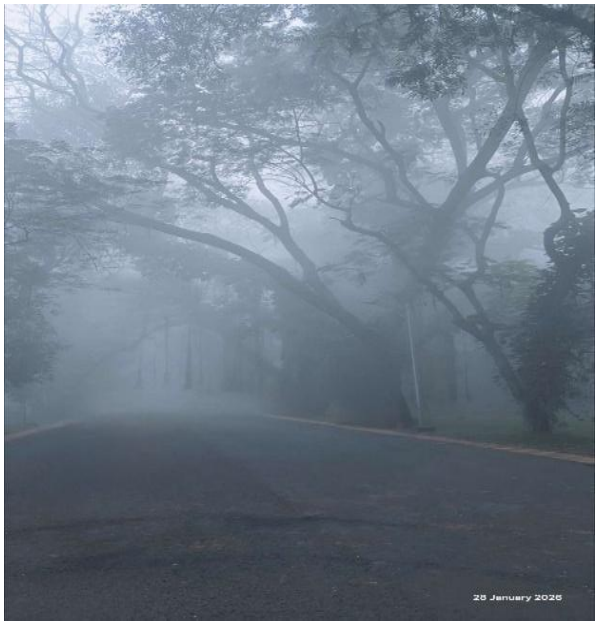

Figure 6.3: Real Time Nosie Image

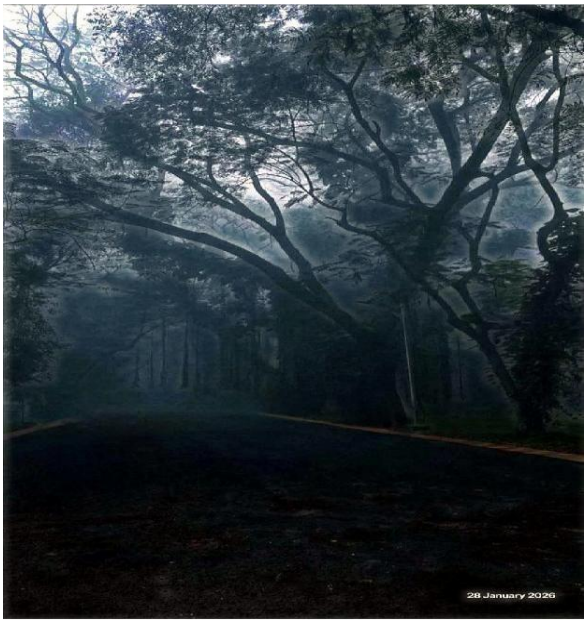

Figure 6.4: Real Time Proposed Method

Tables 2 and 3 present the quantitative evaluation of the three enhancement methods using two widely accepted image quality metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). These metrics provide an objective way to measure how closely the enhanced images match the ground truth images.

In Table 2 (Image 1), Method 1 (NLM + CLAHE) achieves a PSNR of 15.3567 and an SSIM of 0.5174, indicating moderate noise reduction but limited structural preservation. Method 2 (k-SVD + CLAHE) improves the performance, achieving a PSNR of 16.7938 and SSIM of 0.5913. This shows that k-SVD provides better reconstruction and preserves more image details compared to NLM. Method 3 (DCP + CLAHE) significantly outperforms the other methods, with a PSNR of 19.8245 and SSIM of 0.7914. These higher values indicate superior enhancement, better contrast restoration, and improved structural similarity to the ground truth image.

A similar trend is observed in Table 3 (Image 2). Method 1 records the lowest performance (PSNR: 16.0353, SSIM: 0.5006), followed by Method 2 (PSNR: 17.6712, SSIM: 0.5831). Method 3 again achieves the best results with a PSNR of 18.9542 and SSIM of 0.7483. Although slightly lower than Image 1, these values still demonstrate strong performance.

Overall, the results confirm that DCP combined with CLAHE consistently provides the highest image quality, followed by k-SVD and NLM. This validates the effectiveness of the proposed approach in both visual and quantitative terms.

Method	**PSNR**	**SSIM**
Method 1(NLM + CLAHE)	15.3567	0.5174
Method 2(k-SVD + CLAHE)	16.7938	0.5913
Method 3(DCP + CLAHE)	**19.8245**	**0.7914**

Table 2: Quantitative analysis of image 1

Method	**PSNR**	**SSIM**
Method 1(NLM + CLAHE)	16.0353	0.5006
Method 2(k-SVD + CLAHE)	17.6712	0.5831
Method 3(DCP + CLAHE)	**18.9542**	**0.7483**

Table 3: Quantitative analysis of image 2

From the quantitative analysis presented in Table 2 and Table 3, it is clearly observed that the performance of the image enhancement methods improves progressively across the different techniques. The NLM with CLAHE method provides an initial level of enhancement with moderate PSNR and SSIM values. The K-SVD with CLAHE method further improves these metric values, indicating better reconstruction accuracy and enhanced structural similarity when compared to NLM with CLAHE. This improvement is due to the effective sparse representation capability of the K-SVD algorithm, which enhances detail preservation and reduces distortion. However, the proposed DCP with CLAHE method achieves the highest PSNR and SSIM values among all the methods. This indicates that the enhanced images produced by this method are closest to the ground truth images in terms of both pixel-level accuracy and structural consistency.

The consistent improvement in metric values from NLM with CLAHE to K-SVD with CLAHE and finally to DCP with CLAHE demonstrates the superiority of the proposed approach. Based on this comparative analysis, the DCP with CLAHE method is selected as the proposed work in this study. The higher PSNR values confirm better reconstruction quality, while the higher SSIM values indicate improved preservation of structural features such as edges and textures. Therefore, the proposed DCP with CLAHE method provides the most effective solution for enhancing images degraded by fog, smoke, and sand dust, as it achieves the best balance between noise reduction, contrast enhancement, and structural preservation.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

In conclusion, the proposed visibility restoration framework effectively addresses image degradation caused by atmospheric conditions such as fog, smoke, and sand dust by utilizing the Dark Channel Prior (DCP) combined with the Contrast Limited Adaptive Histogram Equalization (CLAHE) method, which accurately estimates the dark channel, atmospheric light, and transmission map to recover scene radiance and significantly improve image visibility. The refinement of the transmission map ensures smooth transitions while preserving important structural and edge details, thereby reducing artifacts and maintaining image quality. Furthermore, the application of CLAHE enhances local contrast, brightness, and color distribution, resulting in clearer, sharper, and more visually informative images. The experimental results demonstrate that the proposed method outperforms other techniques such as NLM and K-SVD with CLAHE, achieving higher PSNR and SSIM values, which indicate better noise reduction, contrast enhancement, and structural preservation. Additionally, the method is computationally efficient and simple to implement, making it suitable for real-time and practical applications in fields such as surveillance, transportation, remote sensing, and everyday imaging scenarios, where improved visibility and accurate image interpretation are essential.

7.2 Future Scope

The proposed image enhancement framework can be further improved by integrating deep learning techniques for more adaptive and accurate dehazing. Future work may focus on reducing computational complexity to enable real-time processing on embedded systems. The method can also be extended to handle video sequences for continuous visibility restoration. Incorporating color correction and artifact reduction techniques can further enhance visual quality. This approach can be applied in advanced applications such as autonomous vehicles, medical imaging, and smart surveillance systems.

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